In class lab #6 for Numerical Analysis

Your name: _____

scan and submit before 23:00

Part I: polynomial interpolation (30 points)

Problem 1. (30 points)

(a) Find a polynomial $P_3(x) = c_0 + c_1x + c_2x^2 + c_3x^3$ of degree 3 or less whose graph passes through the points $(0,0), (1,1), (2,2), (3,7)$ by solving a linear system like $X\vec{c} = \vec{y}$.

(b) Find two other polynomials (of any degree) that pass through these four points.

(c) Decide whether there exists a polynomial P(x) of degree 3 or less whose graph passes through the points $(0,0), (1,1), (2,2), (3,7), (4,2)$.

Part II: Lagrange Interpolation (80 points)

Generally $P_{n-1}(x)$: Suppose that we are given n points $(x_1, y_1), (x_2, y_2), \cdots, (x_n, y_n)$. Then the Lagrange interpolating polynomial is

$$P_{n-1}(x) = y_1 \cdot L_1(x) + y_2 \cdot L_2(x) + \cdots + y_n \cdot L_n(x),$$

where

$$\begin{aligned} L_i(x) &= \prod_{j=1, j\neq i}^{n} \left(\frac{x - x_j}{x_i - x_j} \right) \\ &= \frac{(x - x_1)(x - x_2) \cdots (x - x_{i-1})(x - x_{i+1}) \cdots (x - x_n)}{(x_i - x_1)(x_i - x_2) \cdots (x_i - x_{i-1})(x_i - x_{i+1}) \cdots (x_i - x_n)}. \end{aligned}$$

Problem 2. Step 1. (5 points) Click on New to create a new document and type the code below. Once you are done, save it by the name `evaluate_lagrange.m`.:

```
% Y=evaluate_lagrange(x,y,X)
%
% Input:
% x : vector with n+1 entries (points x_i)
% y : vector with n+1 entries (points y_i)
% X : points where we want to evaluate
% Output:
% Y : values of the interpolant P at the points X
function Y=evaluate_lagrange(x,y,X)
Y=0;
n=length(x)-1;
for k=1:n+1
c=y(k);
for j=[1:k-1 , k+1:n+1] % list of 1 to n+1 without k
c=c.*(X-x(j))./(x(k)-x(j));
end
Y=Y+c;
end
return
```

Step 2 (15 points) Use `evaluate_lagrange.m` to get a polynomial of degree 2, $P_3(x)$, that passes through the points $(0, 0), (1, 1), (2, 2), (3, 7)$ and evaluate the value $P_3(1.5)$. Write down the code and its result.

Problem 3. (30 points) Use `evaluate_lagrange.m` to get the Lagrangian interpolation polynomial $P_2(x)$ for $f(x) = \sqrt{x}$ at the following four sets of points respectively. Compute $P_2(x)$ and the error with $f(x)$, at points $x = 5, 50, 115, 185$.

(a) $(1, f(1)), (4, f(4)), (9, f(9))$.

(b) $(36, f(36)), (49, f(49)), (64, f(64))$.

(c) $(100, f(100)), (121, f(121)), (144, f(144))$.

(d) $(169, f(169)), (196, f(196)), (225, f(225))$.

Report the results and state what you observe.

Problem 4. (20 points) Let $f(x) = 1/(1 + x^2)$.

(a) Use Matlab to plot $f(x)$ in $[-5, 5]$.

(b) Use Matlab to plot the Lagrangian interpolation polynomial for $f(x)$ at n evenly spaced points in $[-5, 5]$, $n = 3, 5, 10, 15, 20$. Show the figures and report what you observe.